

Valuation of Options - part 3

Quantitative Finance

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Assumptions Black-Scholes market (recap)

In class we will almost exclusively work with the 'Black-Scholes market'.

Assumptions on price processes:

Asset price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad \text{var}(W_1) = 1$$

Money Market Account:

$$dB_t = rB_t dt, \quad B_0 = 1$$

Assumptions on market:

- frictionless trading
 - no transaction costs
 - trading in continuous-time is possible
 - no restrictions on short sales and fractional positions
- no credit/counterparty risk: borrowing rate = lending rate

- we are interested in the price C_t , for $t \in [0, T)$, of (European) option with payoff $h(S_T)$ at expiration date/maturity T
- we will discuss three methods to determine fair price (using continuous-time model for financial market):
 - Black-Scholes Partial Differential Equation
 - risk-neutral pricing
 - pricing kernel

These slides discuss risk-neutral pricing and the pricing kernel approach.

The logic of risk-neutral pricing:

- ➊ **First Fundamental Theorem:** no arbitrage $\Leftrightarrow$ there is a measure $\mathbb{Q} \sim \mathbb{P}$ under which *discounted* prices are martingales.
- ➋ To *use* it we must actually find $\mathbb{Q}$ - this is what **Girsanov's** theorem delivers (an intermezzo).
- ➌ **Apply it:** under $\mathbb{Q}$ the stock earns r , and the option price is a discounted $\mathbb{Q}$ -expectation. Worked examples (digital put, Philips put).
- ➍ An equivalent encoding: the **pricing kernel**.
- ➎ When is $\mathbb{Q}$ *unique*? The **Second Fundamental Theorem** (completeness).

Section 1

Risk-neutral pricing

In part 1 we priced options in the one-period toy model two ways: by *replication*, and by *risk-neutral* pricing under a measure $\mathbb{Q}$ that made S/B a martingale. There we proved the First Fundamental Theorem of asset pricing.

The same ideas carry over to continuous time - only the machinery is heavier. We now

- state the continuous-time First Fundamental Theorem;
- *find* the risk-neutral measure $\mathbb{Q}$ (via Girsanov);
- use it to price European options.

The First Fundamental Theorem of asset pricing

Theorem

Under regularity conditions: absence of arbitrage holds if and only if, for some numéraire N , there exists a probability measure

$\mathbb{Q} = \mathbb{Q}_N$ such that:

- (1) $\mathbb{Q}$ is equivalent to $\mathbb{P}$;
 - i.e. $\mathbb{Q}(A) > 0$ iff $\mathbb{P}(A) > 0$ and $\mathbb{Q}(A) = 0$ iff $\mathbb{P}(A) = 0$;*
- (2) for any asset A in the market: the discounted price process A/N is a $\mathbb{Q}$ -martingale, i.e.*

$$\frac{A_t}{N_t} = \mathbb{E}_{\mathbb{Q}} \left[\frac{A_{t+h}}{N_{t+h}} \mid \mathcal{F}_t \right] = \mathbb{E}_{\mathbb{Q}} \left[\frac{A_T}{N_T} \mid \mathcal{F}_t \right].$$

Remarks:

- any traded asset A with $\mathbb{P}(A_t > 0) = 1$ for all t can be used as numéraire
- typically the Money Market Account B is used as numéraire

We will only give a full proof for the Black-Scholes market. To prepare for the proof we first notice the following:

- Consider Black-Scholes market
- Use $N = B$
- If V is self-financing portfolio then

$$\begin{aligned}d\frac{V_t}{B_t} &= \frac{1}{B_t} dV_t - r\frac{V_t}{B_t} dt = \frac{1}{B_t}(\phi_t dS_t + \psi_t dB_t) - r\frac{V_t}{B_t} dt \\ &= \dots = \phi_t d\frac{S_t}{B_t}\end{aligned}$$

Hence: if S/B is $\mathbb{Q}$ -martingale, then V/B is $\mathbb{Q}$ -martingale as well!

First Fundamental Theorem of asset pricing

Proof.

' $\Leftarrow$ ' (general proof; not only for Black-Scholes market)

- let V be the value process corresponding to a self-financing trading strategy
- suppose V satisfies $\mathbb{P}(V_T < 0) = 0$ and $\mathbb{P}(V_T > 0) > 0$
 - if $\mathbb{P}(V_0 \leq 0) = 1$ we would have an arbitrage opportunity
- equivalence yields: $\mathbb{Q}(V_T < 0) = 0$ and $\mathbb{Q}(V_T > 0) > 0$
- we have (use previous slide!)

$$V_0 = N_0 \mathbb{E}_{\mathbb{Q}} \left[\frac{V_T}{N_T} \mid \mathcal{F}_0 \right] = N_0 \mathbb{E}_{\mathbb{Q}} \left[\frac{V_T}{N_T} \right] > 0$$

- consequently $\mathbb{P}(V_0 > 0) > 0 \implies$ strategy does not yield arbitrage opportunity

' $\Rightarrow$ ' We do not prove this implication in general. For the Black-Scholes market you will see an 'explicit' construction. □

Using the first fundamental theorem to price options

- (a) Construct model for basic assets (for example, S and B).
- (b) Apply FFT: If your model does not allow for arbitrage opportunities, FFT says that you should be able to find $\mathbb{Q} \sim \mathbb{P}$ such that S/B is $\mathbb{Q}$ -martingale (assuming that you use B as numéraire, note that B/B then yields a martingale). Use this to determine $\mathbb{Q}$. We will (typically) find unique solution.
- (c) Now introduce an option (on S) to the market. We know its payoff $h(S_T)$ at maturity and want to determine its no-arbitrage price C_t for $t < T$.
- (d) Apply FFT: B/N , S/N , and C/N should be $\mathbb{Q}$ -martingales. From step (b) you already obtained unique $\mathbb{Q}$. This implies that we must set price of option C_t , for $t \in [0, T)$, via

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[\frac{C_T}{N_T} \mid \mathcal{F}_t \right] = N_t \mathbb{E}_{\mathbb{Q}} \left[\frac{h(S_T)}{N_T} \mid \mathcal{F}_t \right].$$

Using the first fundamental theorem to price options

- price processes basic assets in standard Black-Scholes market given; in particular:

$$S_T = S_t \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) (T - t) + \sigma (W_T^{\mathbb{P}} - W_t^{\mathbb{P}}) \right),$$

where $W^{\mathbb{P}}$ is a standard $\mathbb{P}$ -Brownian motion

- use these to solve for $\mathbb{Q}$
 - how?**
- want to price European option with payoff $C_T = h(S_T)$:

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[\frac{C_T}{N_T} \mid \mathcal{F}_t \right] = N_t \mathbb{E}_{\mathbb{Q}} \left[\frac{h(S_T)}{N_T} \mid \mathcal{F}_t \right]$$

- in case $N_t = B_t = \exp(rt)$:

$$C_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} [C_T \mid \mathcal{F}_t] = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} [h(S_T) \mid \mathcal{F}_t]$$

- $\implies$ **we do not need to know $\mathbb{Q}$, but 'only' need to know the distribution of $W^{\mathbb{P}}$, and hence S_T , under $\mathbb{Q}$!**

Theorem (Girsanov)

Suppose the process $W^{\mathbb{P}}$ is a standard Brownian motion under $\mathbb{P}$ and let $T < \infty$. Consider a process $(\gamma_t)_{t \geq 0}$ adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$ generated by $W^{\mathbb{P}}$ and satisfying appropriate integrability conditions. Define $\mathbb{Q}$ by, for events $A \in \mathcal{F}_T$,

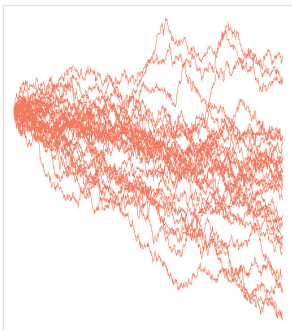
$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}} \left[1_A \exp \left(-\frac{1}{2} \int_0^T \gamma_t^2 dt - \int_0^T \gamma_t dW_t^{\mathbb{P}} \right) \right].$$

Then we have:

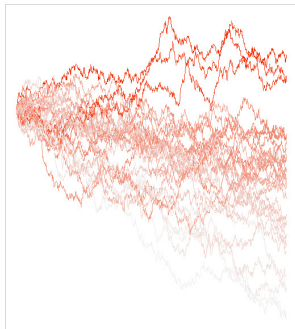
- $\mathbb{Q}$ is a probability measure equivalent to $\mathbb{P}$,
- and the process $W^{\mathbb{Q}}$ defined by

$$W_0^{\mathbb{Q}} = 0, \quad dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}},$$

or equivalently $W_t^{\mathbb{Q}} = \int_0^t \gamma_u du + W_t^{\mathbb{P}}$, is a standard Brownian motion under $\mathbb{Q}$.



30 paths of a Brownian motion with negative drift



The same paths reweighted according to the Girsanov formula

- source: Wikipedia
- under $\mathbb{Q}$ and $\mathbb{P}$ same sample paths possible (equivalence), but with different likelihoods
- a sketch of proof is given in the appendix

- Girsanov: for

$$W_0^{\mathbb{Q}} = 0, \quad dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}} \quad (W^{\mathbb{P}} \text{ standard BM under } \mathbb{P})$$

we can find $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that $W^{\mathbb{Q}}$ is standard Brownian motion under $\mathbb{Q}$

- Let $0 < a \neq 1$. Suppose that we can find probability measure $\mathbb{Q}$ such that

$$Z_0^{\mathbb{Q}} = 0, \quad dZ_t^{\mathbb{Q}} = \gamma_t dt + a dW_t^{\mathbb{P}} \quad (W^{\mathbb{P}} \text{ standard BM under } \mathbb{P}),$$

is standard Brownian motion under $\mathbb{Q}$. Then $\mathbb{Q}$ is *not* equivalent to $\mathbb{P}$.

Recall the First Fundamental Theorem (stated earlier): for a numéraire N , absence of arbitrage means there is a measure $\mathbb{Q} \sim \mathbb{P}$ under which every discounted price A/N is a martingale.

Plan. In the Black-Scholes market we now *construct* this $\mathbb{Q}$ explicitly - using Girsanov - and then read off option prices as discounted $\mathbb{Q}$ -expectations.

(We use the money-market account B as numéraire; any asset that stays strictly positive would also do.)

Using FFT and Girsanov: setup

- setup: standard Black-Scholes market ($W = W^{\mathbb{P}}$ is standard Brownian motion):

$$dB_t = rB_t dt, \quad dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad S_0 > 0, \quad B_0 = 1$$

- use B as numéraire
- Easy exercise (Itô) to show that

$$d\frac{S_t}{B_t} = (\mu - r)\frac{S_t}{B_t} dt + \sigma\frac{S_t}{B_t} dW_t^{\mathbb{P}}$$

- FFT $\implies$ we need to find $\mathbb{Q}$ such that S_t/B_t is $\mathbb{Q}$ -martingale
 - note that B/B is automatically a martingale

Using FFT and Girsanov: finding $\mathbb{Q}$

$$d \frac{S_t}{B_t} = (\mu - r) \frac{S_t}{B_t} dt + \sigma \frac{S_t}{B_t} dW_t^{\mathbb{P}}$$

- for

$$dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}}$$

we have $dW_t^{\mathbb{P}} = -\gamma_t dt + dW_t^{\mathbb{Q}}$ which yields

$$d \frac{S_t}{B_t} = ((\mu - r) - \sigma \gamma_t) \frac{S_t}{B_t} dt + \sigma \frac{S_t}{B_t} dW_t^{\mathbb{Q}}$$

- application of Girsanov's theorem with

$$\gamma_t = \frac{\mu - r}{\sigma} \quad \text{market price of risk}$$

yields $\mathbb{Q} \sim \mathbb{P}$ such that $W^{\mathbb{Q}}$ is a $\mathbb{Q}$ -standard Brownian motion

- we have

$$d \frac{S_t}{B_t} = \sigma \frac{S_t}{B_t} dW_t^{\mathbb{Q}}$$

so S/B is martingale under $\mathbb{Q}$ (not under $\mathbb{P}$!)

From the binomial model to the market price of risk

In part 1 the risk-neutral probability q tilted the real-world p just enough to make S/B a martingale. In continuous time Girsanov does the same job continuously, through the **market price of risk**

$$\gamma = \frac{\mu - r}{\sigma}.$$

- $\mu - r$ is the stock's *excess return* over the riskless rate; dividing by σ gives excess return *per unit of risk* (a Sharpe ratio);
- Girsanov removes exactly this drift, $dW_t^{\mathbb{Q}} = \gamma dt + dW_t^{\mathbb{P}}$, turning the stock's drift from μ into r ;
- discrete analogue (part 1): $q = \frac{B_1/B_0 - s_d/S_0}{s_u/S_0 - s_d/S_0}$ is the same tilt, one period at a time.

The larger the risk premium $\mu - r$ (or the smaller the risk σ), the larger the adjustment γ .

Remark:

Under $\mathbb{Q}$ we have

$$d \frac{S_t}{B_t} = \sigma \frac{S_t}{B_t} dW_t^{\mathbb{Q}}$$

$\implies$ (use Itô)

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

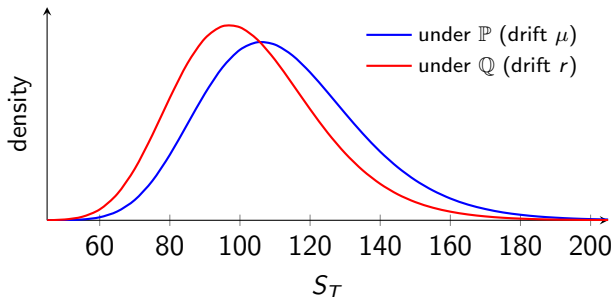
- under $\mathbb{P}$ we have

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}$$

- under $\mathbb{Q}$ stock has same drift as bond, but with risk $\implies$
risk-neutral (world)

The real-world and risk-neutral laws of S_T

The change of measure $\mathbb{P} \rightarrow \mathbb{Q}$ shifts the stock's drift from μ to r but leaves σ unchanged. So S_T is lognormal under both, with the *same* spread but a different centre:



Here $S_0 = 100$, $\sigma = 20\%$, $T = 1$, $\mu = 12\%$, $r = 3\%$; then $\mathbb{E}_{\mathbb{P}}[S_T] = 112.8$ but $\mathbb{E}_{\mathbb{Q}}[S_T] = 103.0$. Same outcomes (equivalent measures!), different likelihoods.

Worked example: under $\mathbb{Q}$ the stock earns the riskless rate

Under $\mathbb{Q}$, $S_T = S_t e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})}$ with $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \sim N(0, T-t)$. Using $\mathbb{E}[e^{\sigma Z'}] = e^{\frac{1}{2}\sigma^2(T-t)}$ for $Z' \sim N(0, T-t)$:

$$\mathbb{E}_{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = S_t e^{(r - \frac{1}{2}\sigma^2)(T-t)} e^{\frac{1}{2}\sigma^2(T-t)} = S_t e^{r(T-t)}.$$

Equivalently $\mathbb{E}_{\mathbb{Q}}[S_T/B_T \mid \mathcal{F}_t] = S_t/B_t$: S/B is a $\mathbb{Q}$ -martingale, as it must be. The same computation under $\mathbb{P}$ gives $\mathbb{E}_{\mathbb{P}}[S_T \mid \mathcal{F}_t] = S_t e^{\mu(T-t)}$ - the stock earns its true drift $\mu > r$, a *risk premium*.

This is the continuous-time version of the binomial identity

$$\frac{1}{B_1} \mathbb{E}_{\mathbb{Q}}[S_1] = S_0 \text{ from part 1.}$$

Using FFT and Girsanov: the pricing formula

- the SDE

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

is GBM

- so, under $\mathbb{Q}$,

$$S_T = S_t e^{(r-0.5\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})}$$

- option with payoff $h(S_T)$
- FFT yields

$$\begin{aligned} C_t &= e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left[h \left(S_t e^{(r-0.5\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})} \right) \mid \mathcal{F}_t \right] \\ &= e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left[h \left(S_t e^{(r-0.5\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})} \right) \right] \end{aligned}$$

(we may drop the conditioning: S_t is known at time t and $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}}$ is independent of $\mathcal{F}_t$)

- using $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \sim \sqrt{T-t}Z$ (under $\mathbb{Q}$) with $Z \sim N(0, 1)$:

$$C_t = e^{-r(T-t)} \int_{-\infty}^{\infty} h \left(S_t e^{(r-0.5\sigma^2)(T-t) + \sigma\sqrt{T-t}z} \right) \phi(z) dz$$

Example - digital put

- digital European put: $h(S_T) = 1\{S_T \leq K\}$; what is the price at $t = 0$?
- we have to calculate

$$C_0 = \exp(-rT) \mathbb{E}_{\mathbb{Q}} [h(S_T)] = \exp(-rT) \mathbb{Q}(S_T \leq K)$$

- using

$$\begin{aligned} S_T &= S_0 \exp\left((r - 0.5\sigma^2)T + \sigma W_T^{\mathbb{Q}}\right) \leq K \\ &\Leftrightarrow \sigma W_T^{\mathbb{Q}} \leq \log \frac{K}{S_0} - (r - 0.5\sigma^2)T \end{aligned}$$

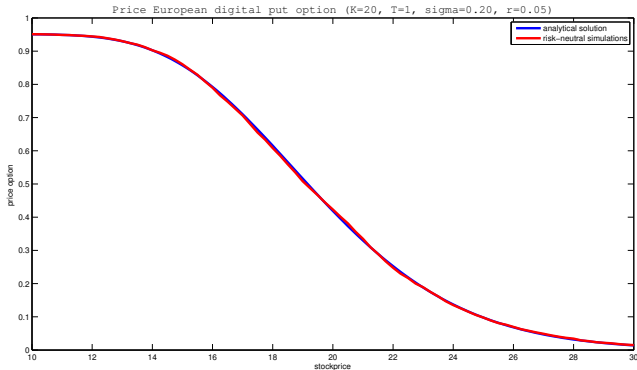
we obtain

$$\begin{aligned} C_0 &= \exp(-rT) \mathbb{Q}\left(\sigma W_T^{\mathbb{Q}} \leq \log \frac{K}{S_0} - (r - 0.5\sigma^2)T\right) \\ &= \exp(-rT) \Phi\left(\frac{\log \frac{K}{S_0} - (r - 0.5\sigma^2)T}{\sigma\sqrt{T}}\right) \end{aligned}$$

This is exactly the price the part-2 PDE approach gives for the digital put - the two methods agree.

Example digital put

The blue line plots C_0 as function of S_0 for $K = 20$, $T = 1$, $\sigma = 20\%$, and $r = 5\%$.



Later on we will discuss how we can use Monte Carlo techniques to obtain a numerical approximation. The red line is such an approximation based on 5,000 simulations.

Quiz

In the Black-Scholes market with $\mu = 10\%$, $r = 2\%$, $\sigma = 25\%$:

- 1 What is the market price of risk γ ?
- 2 Write the SDE for S under the risk-neutral measure $\mathbb{Q}$.
- 3 What is the distribution of $\log S_T$ under $\mathbb{Q}$ (given S_0)?
- 4 Is $\mathbb{E}_{\mathbb{Q}}[S_T]$ larger or smaller than $\mathbb{E}_{\mathbb{P}}[S_T]$?

Example: put on Philips



historical data yields estimated $\sigma = 0.29$

Example: put on Philips

DECEMBER 2021 PRICES - 17/07/20

	SETTL.	LAST	BID	ASK		STRIKE		BID	LAST	SETTL.
+	18.02	-	17.70	18.00	C	25.00	P	0.46	-	0.52
+	13.63	-	13.30	13.60	C	30.00	P	1.12	-	1.17
+	9.75	-	9.45	9.75	C	35.00	P	2.29	-	2.29
+	6.48	-	6.20	6.50	C	40.00	P	4.07	-	4.06
+	3.98	-	3.81	4.00	C	45.00	P	6.60	-	6.60
+	2.24	-	2.14	2.29	C	50.00	P	9.95	-	9.93
+	0.60	-	0.52	0.67	C	60.00	P	18.40	-	18.37



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Help

- $T = 17.5/12$, $r = 0.01$, $K = 45$, $\sigma = 0.29$ and $S_0 = 43.145$ yields 6.70 as B-S price for the put, where $\sigma = 0.29$ is result from estimation on historical data on S .

Food for thought

The risk-neutral measure $\mathbb{Q}$ is *not* the real-world measure: it never describes the actual probabilities of price moves, and under $\mathbb{Q}$ every asset earns only r .

- In what sense, then, is $\mathbb{Q}$ “fictional but useful”?
- Two investors who disagree about the drift μ (hence about $\mathbb{P}$) still agree on the option price. Why does μ drop out?
- If a second, *untraded* source of risk is added, $\mathbb{Q}$ is no longer unique (Second FTAP). What breaks in the replication argument?

Section 2

Pricing Kernels

Motivation

- suppose we work in standard Black-Scholes market with B as numéraire
- Girsanov yields probability measure $\mathbb{Q}$ defined by $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}} 1_A Z_T$ where

$$Z_t = \exp \left(-t(\mu - r)^2 / (2\sigma^2) - ((\mu - r)/\sigma) W_t^{\mathbb{P}} \right)$$

- we know $\mathbb{E}_{\mathbb{Q}}[X] = \mathbb{E}_{\mathbb{P}}[XZ_T]$
- it can be shown that $\mathbb{E}_{\mathbb{Q}}[X|\mathcal{F}_t] = \mathbb{E}_{\mathbb{P}}[X(Z_T/Z_t)|\mathcal{F}_t]$
- we thus have

$$A_t = \mathbb{E}_{\mathbb{Q}} \left[\frac{N_t}{N_T} A_T \mid \mathcal{F}_t \right] = \mathbb{E}_{\mathbb{P}} \left[\frac{N_t}{N_T} \frac{Z_T}{Z_t} A_T \mid \mathcal{F}_t \right]$$

which yields, with $K_t = Z_t/N_t$,

$$K_t A_t = \mathbb{E}_{\mathbb{P}}[K_T A_T \mid \mathcal{F}_t],$$

i.e. KA is a $\mathbb{P}$ -martingale

Here $K = Z/N$ is the discounted likelihood ratio $\frac{1}{N} \frac{d\mathbb{Q}}{d\mathbb{P}}$ - the continuous-time version of the kernel $m = \frac{B_0}{B_1} \frac{d\mathbb{Q}}{d\mathbb{P}}$ from part 1.

The first fundamental theorem of asset pricing (pricing kernel version)

We proved, for the special case of the Black-Scholes market, the following theorem.

Theorem

Absence of arbitrage holds if and only if there exists a process K , with $K_t > 0$, such that for any asset with price A_t the process $K_t A_t$ is a $\mathbb{P}$ -martingale

Application:

- determine K (using your models for the basic assets)
- (no arbitrage) price of a derivative, with payoff C_T , based on these assets is given by

$$C_t = \frac{1}{K_t} \mathbb{E}_{\mathbb{P}}[K_T C_T \mid \mathcal{F}_t]$$

The pricing kernel for the standard Black-Scholes world

- try K of the form $dK_t = \mu_t^K dt + \sigma_t^K dW_t^{\mathbb{P}}$
- Itô's product rule yields

$$\begin{aligned}dK_t B_t &= K_t dB_t + B_t dK_t + d[K, B]_t \\&= (rK_t B_t + B_t \mu_t^K) dt + \dots dW_t^{\mathbb{P}}\end{aligned}$$

$$K_t B_t \text{ martingale} \implies \text{no drift} \implies \mu_t^K = -rK_t$$

- Itô's product rule yields

$$\begin{aligned}dK_t S_t &= K_t dS_t + S_t dK_t + d[S, K]_t \\&= (K_t \mu S_t + S_t \mu_t^K + \sigma \sigma_t^K S_t) dt + \dots dW_t^{\mathbb{P}}\end{aligned}$$

$$K_t S_t \text{ martingale} \implies \text{no drift} \implies \sigma_t^K = -K_t(\mu - r)/\sigma$$

So

$$K_0 = 1, \quad dK_t = -rK_t dt - \frac{\mu - r}{\sigma} K_t dW_t^{\mathbb{P}}$$

determines the pricing kernel process in the B-S model.

The pricing kernel for the standard Black-Scholes world

$$K_0 = 1, \quad dK_t = -rK_t dt - \frac{\mu - r}{\sigma} K_t dW_t^{\mathbb{P}}$$

has as solution

$$K_t = \exp \left\{ \left(-r - 0.5 \left(\frac{\mu - r}{\sigma} \right)^2 \right) t - \frac{\mu - r}{\sigma} W_t^{\mathbb{P}} \right\}$$

Fair price of a derivative, with payoff $C_T = h(S_T)$, based on these assets is given by

$$C_0 = \frac{1}{K_0} \mathbb{E}_{\mathbb{P}}[K_T C_T]$$

- both K_T and S_T (and hence $K_T C_T = K_T h(S_T)$) are functions of $W_T^{\mathbb{P}}$, so the price is the expectation of a function of a $N(0, T)$ random variable
- price seems to depend on μ , however it does not!

Remark

- you can always set $K_0 = 1$
- which yields

$$C_0 = \mathbb{E}_{\mathbb{P}}[K_T C_T]$$

- suppose $C_T = 1\{\omega \in E\}$, i.e. the pay-off is 1 euro in case event E occurs and 0 otherwise
- then

$$C_0 = \mathbb{E}_{\mathbb{P}}[1_E K_T],$$

so $K_T(\omega)$ kind of measures how expensive it is to get a pay-off in state ω

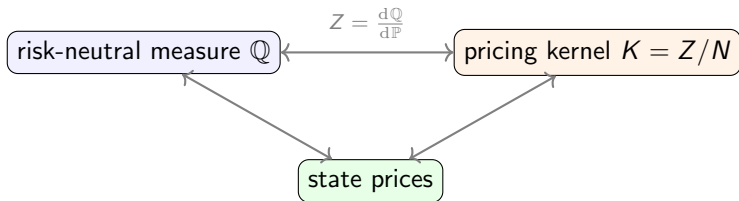
- For the Black-Scholes market we have (assume $\mu > r$)

$$K_T = \exp \left\{ \left(-r - 0.5 \left(\frac{\mu - r}{\sigma} \right)^2 \right) T - \frac{\mu - r}{\sigma} W_T \right\},$$

showing that K_T takes large values for large, negative values of W_T . And such states-of-the-world correspond to small values of $S_T(\omega)$

One object, three encodings

The risk-neutral measure, the pricing kernel, and state prices are three views of the *same* pricing rule.



The same price, three ways:

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[\frac{C_T}{N_T} \mid \mathcal{F}_t \right] = \frac{1}{K_t} \mathbb{E}_{\mathbb{P}} [K_T C_T \mid \mathcal{F}_t] = \int C_T d\pi.$$

In the part-1 toy model these were the measure $\mathbb{Q}$, the kernel m , and the Arrow-Debreu prices π_u, π_d .

Section 3

Second Fundamental Theorem of asset pricing

In our discussion we have assumed that it is possible to find a self-financing portfolio for every option pay-off C_T . This property is referred to as 'completeness'.

Definition

A financial market is said to be **complete** if it is possible to find, for each $T > 0$, and all random variables C_T that are $\mathcal{F}_T$ -measurable, a self-financing portfolio V (satisfying regularity conditions) that satisfies $V_T = C_T$ (a.s.)

Second Fundamental Theorem of asset pricing

Theorem

Under regularity conditions we have: a financial market is complete if and only if there is a *unique* measure $\mathbb{Q} \sim \mathbb{P}$ that makes all deflated asset prices A/N $\mathbb{Q}$ -martingales (where N is the chosen numéraire).

Examples

- For our Black-Scholes market $\mathbb{Q}$ was unique, hence the market is complete.
- Consider the Black-Scholes market. Now we also add an additional risk factor to the market: an independent Brownian motion. This has impact on $\mathcal{F}_T$! And now claims could depend on both Brownian motions. No additional assets are available. This leads to an incomplete market. (Why is $\mathbb{Q}$ not unique?)

Remark In the MSc in QFAS we discuss valuation (and hedging) in incomplete markets (which you often encounter in case there is financial risk as well as actuarial risk).

Three methods, one price

For the European digital put (payoff $1\{S_T \leq K\}$) all three methods give the *same* price:

method	price C_0
Black-Scholes PDE (part 2)	$e^{-rT} \Phi(d_-)$
risk-neutral pricing (part 3)	$e^{-rT} \Phi(d_-)$
pricing kernel (part 3)	$e^{-rT} \Phi(d_-)$

with $d_- = \frac{\log(K/S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$. The PDE, the risk-neutral measure, and the pricing kernel are three routes to the *same* no-arbitrage price - exactly as in the binomial toy model of part 1.

Section 4

Appendix

Appendix: sketch proof of Girsanov (1/2)

- given the integrability conditions, it is easy to show that $\mathbb{Q}$ is probability measure equivalent to $\mathbb{P}$
- to prove that $W^{\mathbb{Q}}$ is a standard Brownian motion under $\mathbb{Q}$ we need to show, for any $0 = t_0 < t_1 < \dots < t_n \leq T$, that

$$(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}, \dots, W_{t_n}^{\mathbb{Q}} - W_{t_{n-1}}^{\mathbb{Q}}) \sim N(0_n, \Sigma)$$

where $\Sigma_{ij} = 0$ for $i \neq j$ and $\Sigma_{ii} = t_i - t_{i-1}$

- we only consider case in which $u \mapsto \gamma_u$ is deterministic and $n = 2$ with $t_2 = T$
- it is sufficient to show (MGF determines the probability distribution)

$$\mathbb{E}_{\mathbb{Q}} \left[e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} \right] = e^{\frac{1}{2}(a_1^2(t_1 - t_0) + a_2^2(t_2 - t_1))}$$

Appendix: sketch proof of Girsanov (2/2)

We have

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}} \left[e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} \right] &= \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} \right] \\&= \mathbb{E}_{\mathbb{P}} \left[e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} e^{-\frac{1}{2} \int_0^T \gamma_u^2 du - \int_0^T \gamma_u dW_u^{\mathbb{P}}} \right] \\&= e^{a_1 \int_{t_0}^{t_1} \gamma_u du + a_2 \int_{t_1}^{t_2} \gamma_u du - \frac{1}{2} \int_{t_0}^{t_2} \gamma_u^2 du} \\&\quad \times \mathbb{E}_{\mathbb{P}} \left[e^{-\int_{t_0}^{t_1} (\gamma_u - a_1) dW_u^{\mathbb{P}} - \int_{t_1}^{t_2} (\gamma_u - a_2) dW_u^{\mathbb{P}}} \right]\end{aligned}$$

Exploiting independent increments and

$\int_a^b f_u dW_u^{\mathbb{P}} \sim N(0, \int_a^b f_u^2 du)$ for deterministic functions f :

$$\begin{aligned}\mathbb{E}_{\mathbb{P}} \left[e^{-\int_{t_0}^{t_1} (\gamma_u - a_1) dW_u^{\mathbb{P}} - \int_{t_1}^{t_2} (\gamma_u - a_2) dW_u^{\mathbb{P}}} \right] \\&= \mathbb{E}_{\mathbb{P}} \left[e^{-\int_{t_0}^{t_1} (\gamma_u - a_1) dW_u^{\mathbb{P}}} \right] \mathbb{E}_{\mathbb{P}} \left[e^{-\int_{t_1}^{t_2} (\gamma_u - a_2) dW_u^{\mathbb{P}}} \right] \\&= e^{\frac{1}{2} \int_{t_0}^{t_1} (\gamma_u - a_1)^2 du} e^{\frac{1}{2} \int_{t_1}^{t_2} (\gamma_u - a_2)^2 du} = e^{\frac{1}{2} \int_{t_0}^{t_2} \gamma_u^2 du - a_1 \int_{t_0}^{t_1} \gamma_u du - a_2 \int_{t_1}^{t_2} \gamma_u du} \\&\quad \times e^{\frac{1}{2} (a_1^2(t_1 - t_0) + a_2^2(t_2 - t_1))} \text{ which yields the result}\end{aligned}$$